Student: Ben Stone

Course: ECE 6210

Subject: Master's Project, Schroeder Reverb

Date: April 20, 2024

1 Introduction

This project is intended to explore the implementation and analysis of the Schroeder Reverb algorithm, a foundational technique in digital audio processing used to simulate acoustic reverberation in audio signals. The focus of the project is to demonstrate the practical application of this algorithm through two distinct configurations, each leveraging a combination of all-pass and comb filters to achieve the desired reverberative effects.

The Schroeder Reverb algorithm is critical in the field of DSP for its efficiency and quality of the reverb it produces, making it a staple in various audio applications such as music production, sound design, and virtual reality environments. By employing multiple all-pass filters in series to create diffuse reflections and multiple comb filters in parallel to simulate echo density, the Schroeder Reverb enhances the naturalness and depth of audio signals.

2 Theory

The Schroeder Reverb algorithm at its most basic is achieved by combining multiple All-pass filters in series and multiple comb filters in parallel. There are multiple ways to connect these filters together and achieve the reverb algorithm for my project I will be demonstrating two of these setups.

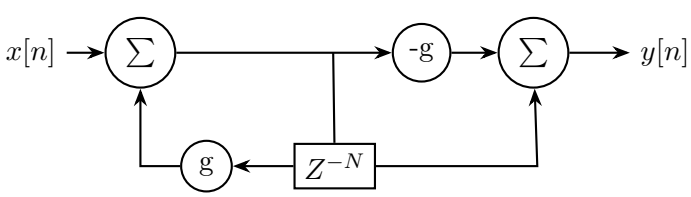
2.1 All-Pass Filter

The all-pass filters provide "colorless" high-density echoes in the late impulse response of the reverberator. These all-pass filters may also be referred to as diffusers. While all-pass filters are "colorless" in theory, perceptually, their impulse responses are only colorless when they are extremely short (less than 10 ms or so). Longer all-pass impulse responses sound similar to feedback comb-filters. For steady-state tones, however, such as sinusoids, the all-pass property gives the same gain at every frequency, unlike comb filters.

The all-pass filter is implemented using the algorithm

$$AP_N^g \triangleq \frac{-g + z^{-N}}{1 - gz^{-N}} \tag{1}$$

The response for this equation is drawn below

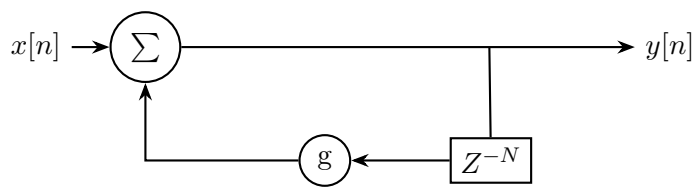


2.2 Comb Filter

The comb filter is implemented using the algorithm

$$FBCF_N^g \triangleq \frac{1}{1 - gz^{-N}} \tag{2}$$

The response for this equation is drawn below



Schroeder suggests a progression of all-pass delay-line lengths close to

$$M_i T \approx \frac{100\,\text{ms}}{3^i}, \quad i = 0, 1, 2, 3, 4 \tag{3}$$

and chosen to be mutually prime (no common factors). The 100 ms value was chosen so that when $g = 0.7$ in Eq. 1, the time to decay 60 dB (t_{60}) would be 2 seconds. Thus, for $i = 0$, $t_{60} \approx 2$, and each successive allpass has an impulse-response duration that is about a third of the previous one.

2.3 Type I

The first setup type is to connect 4 comb filters in parallel using the input signal as input and then summing together the output, this output is then put into 3 all-pass filters in series resulting in the signal output with reverberation and the negated reverberation.

The reverberator of Fig. 1 is thought to have been used in John Chowning's often-heard FM-brass canon sound examples it is designed for a 25 kHz sampling rate. Its "mixing matrix" is simply a negation of the right stereo channel.

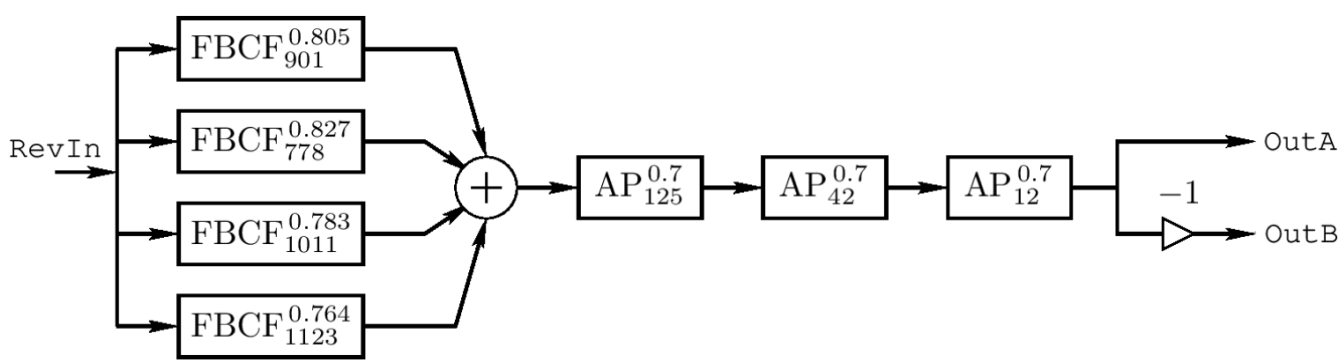


Figure 1: Type I Schroeder reverberator SATREV by Prof. John Chowning at CCRMA (drawn from a 1971 MUS10 software listing).

2.4 Type II

The second setup type is the reverse of the first, I start by putting the input signal through 3 All-Pass, this signal is then used for each of the comb filters in parallel, and each of these inputs are put into a mixing matrix resulting in 4 output signals.

The mixing matrix shown above is expressed as

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ -1 & -1 & -1 & -1 \\ -1 & 1 & -1 & 1 \\ 1 & -1 & 1 & -1 \end{bmatrix}$$

This matrix can be efficiently implemented using four adders and two negations:

$$s_1 = x_1 + x_3 \tag{4}$$
$$s_2 = x_2 + x_4 \tag{5}$$
$$\texttt{OutA} = s_1 + s_2 \tag{6}$$
$$\texttt{OutB} = -\texttt{OutA} \tag{7}$$
$$\texttt{OutC} = -\texttt{OutD} \tag{8}$$
$$\texttt{OutD} = s_1 - s_2 \tag{9}$$

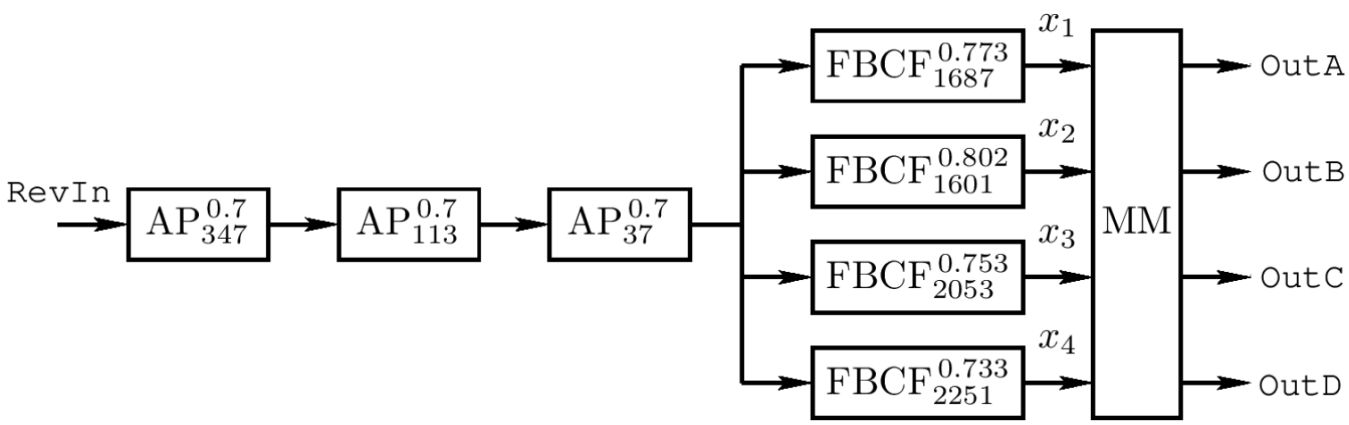


Figure 2: Type II Schroeder reverberator called JCREV developed by Prof. John Chowning, founding director of CCRMA (drawn from a 1972 MUS10 software listing, where MUS10 was an "acoustic compiler" language descended from Music V

3 Results

Provided below is a recording of each of the different audio responses utilizing a portion of the "We choose to go to the Moon" speach, officially titled the address at Rice University on the nation's space effort, given September 12, 1962, by United States President John F. Kennedy to further inform the public about his plan to land a man on the Moon before 1970.

Original Audio
Type I OutA
Type I OutB
Type II OutA
Type II OutB
Type II OutC
Type II OutD

4 Discussion and Conclusions

Considering the output of both Type I and Type II configurations, it's evident that each has a different effect on the original audio signal. Type I exhibits a lighter, higher sound/echo, which might find application in certain artistic contexts, this can be seen in Fig. 3 and 4). Alternatively, Type II presents a slightly louder and more pronounced reverberation, likely attributed to its unique signal processing chain as shown in Fig. 5 to 8.

Interestingly, both configurations extend the duration of the original audio signal, creating a prolonged echo that persists slightly after the source audio ends/pauses. This effect, though subtle, contributes to a sense of spatial immersion and auditory continuity, especially in applications where seamless transitions between audio elements are paramount.
Additionally, it is interesting to note that if the audio has a lot of percussion or sudden sounds both implementations have a poor audio quality when the echoed percussion covers/distorts the other audio. This is even apparent in the included audio, when President Kennedy speaks louder or more suddenly there is a light ringing in the echo of the audio. The Type I design has a lower ring and the Type II has a higher pitch ring.

The choice between Type I and Type II reverberators depends on the specific requirements of the audio project.

5 Resources

https://ccrma.stanford.edu/~jos/pasp/Schroeder_Reverberators.html
https://ccrma.stanford.edu/~jos/pasp/Example_Schroeder_Reverberators.html
https://github.com/Ben1219925/Projects/tree/8bc4b848c3fac26ffa81eed8631c36d9cfee9620/Schroeder%20Reverberator

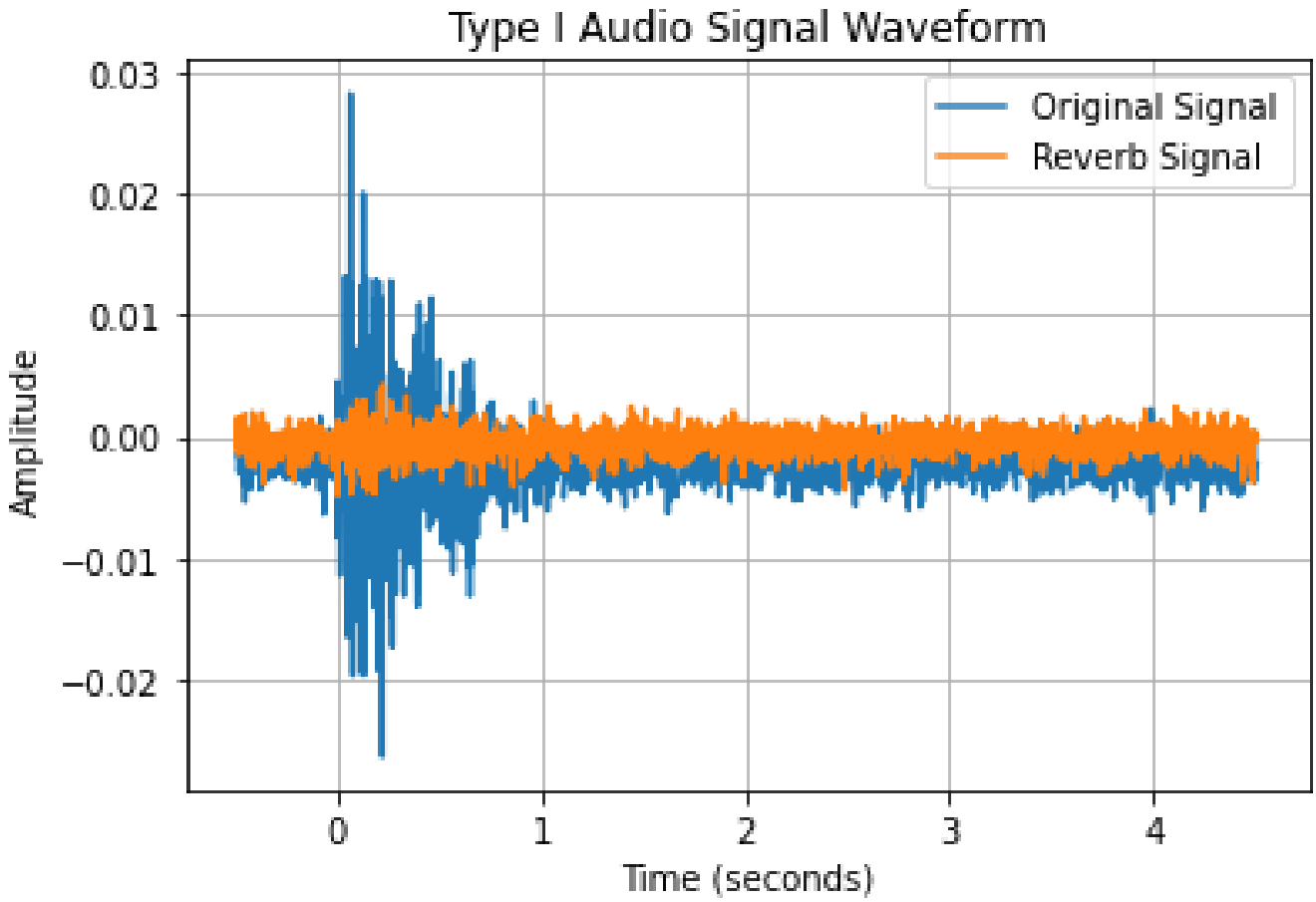


Figure 3: Type I OutA reverb vs. original Drum Darkness Sound effect

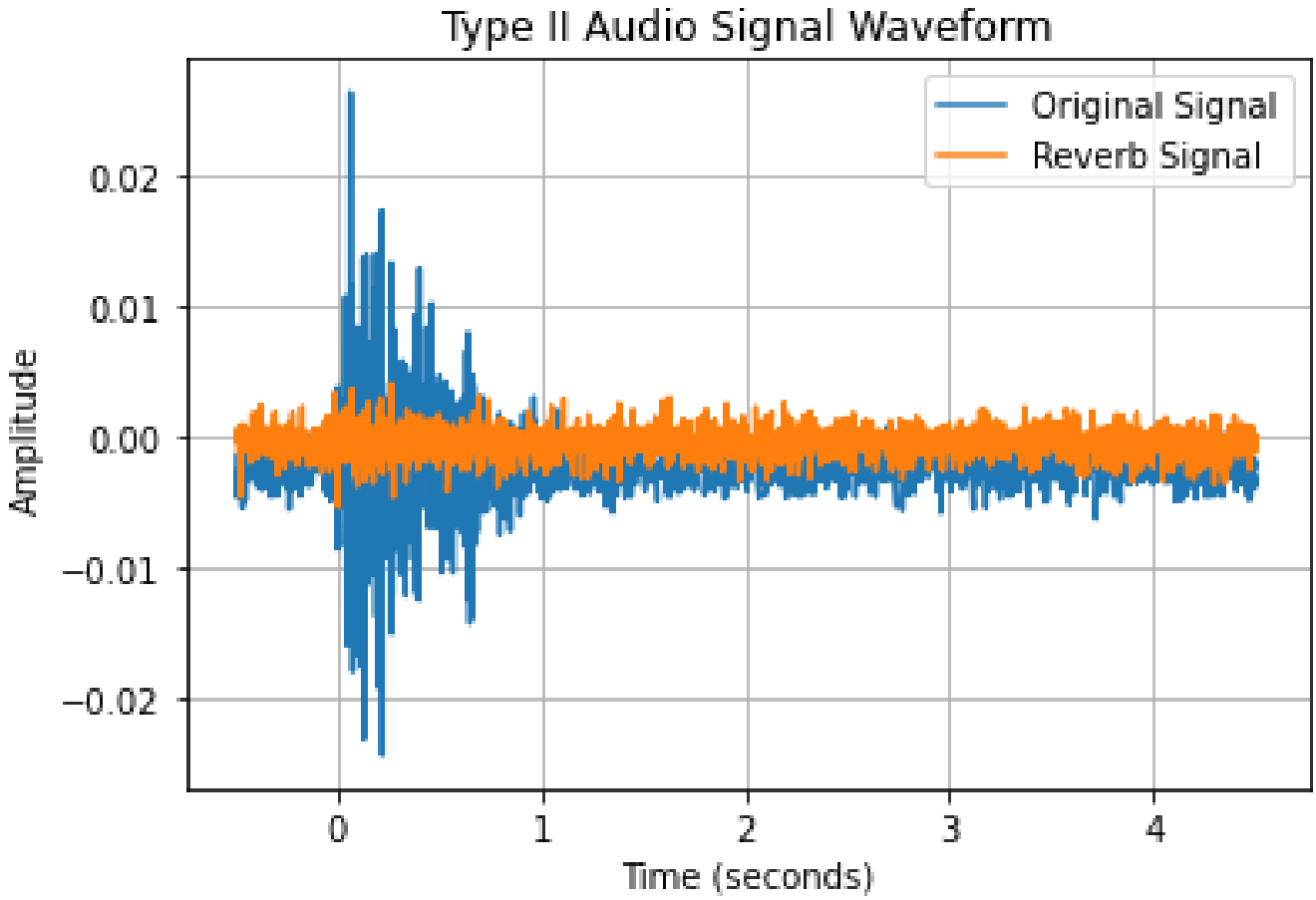


Figure 4: Type I OutB reverb vs. original Drum Darkness Sound effect

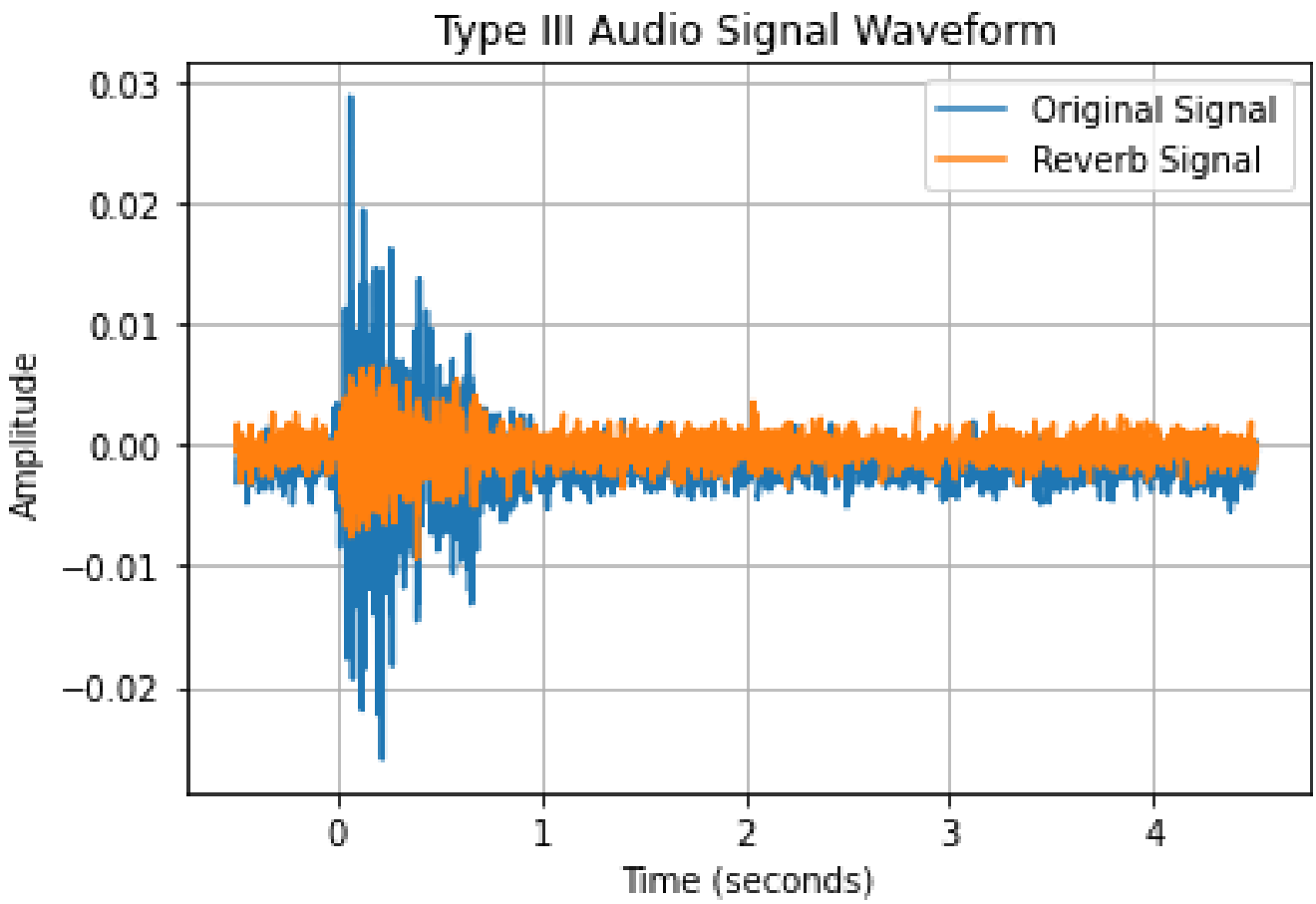


Figure 5: Type II OutA reverb vs. original Drum Darkness Sound effect

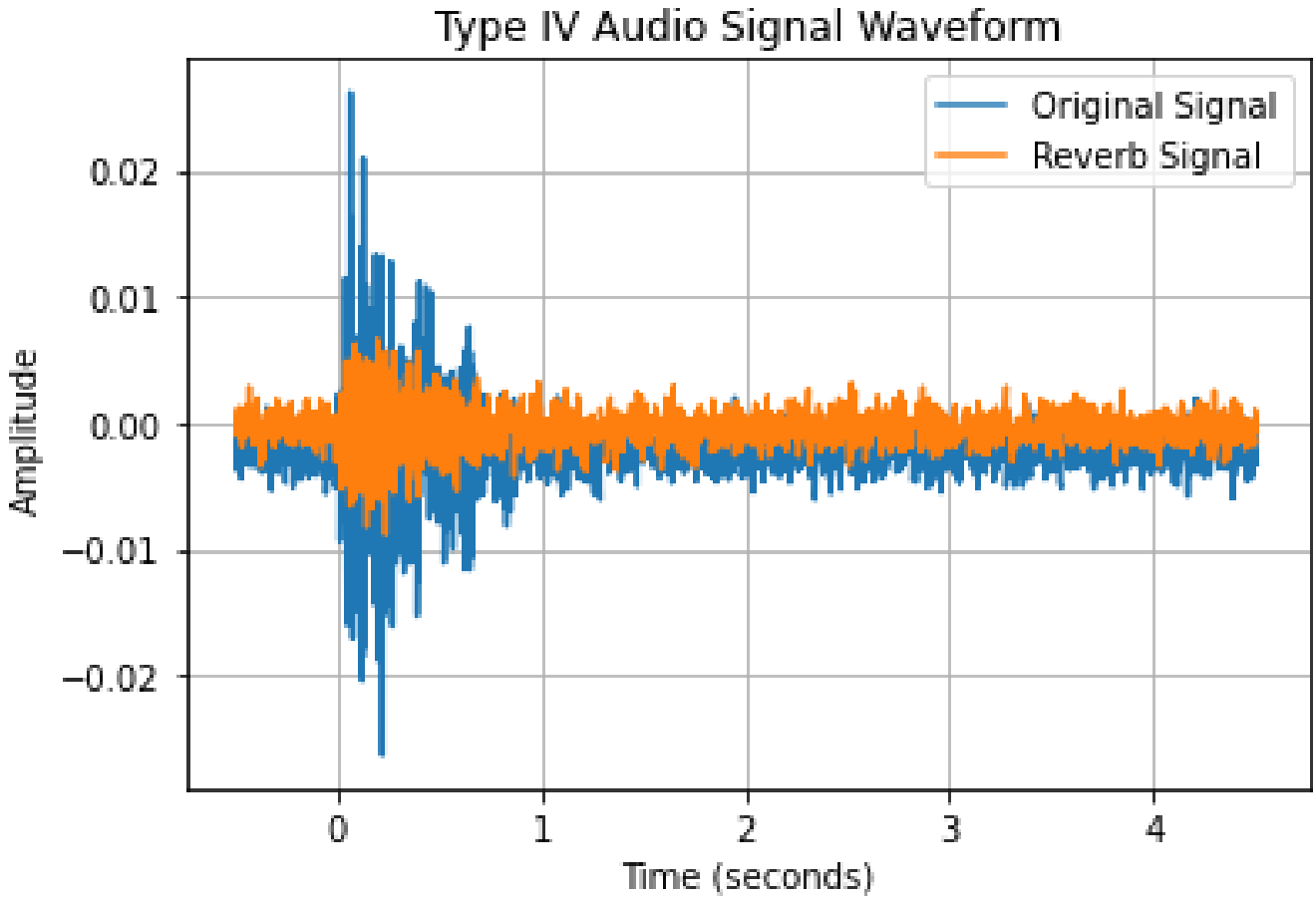


Figure 6: Type II OutB reverb vs. original Drum Darkness Sound effect

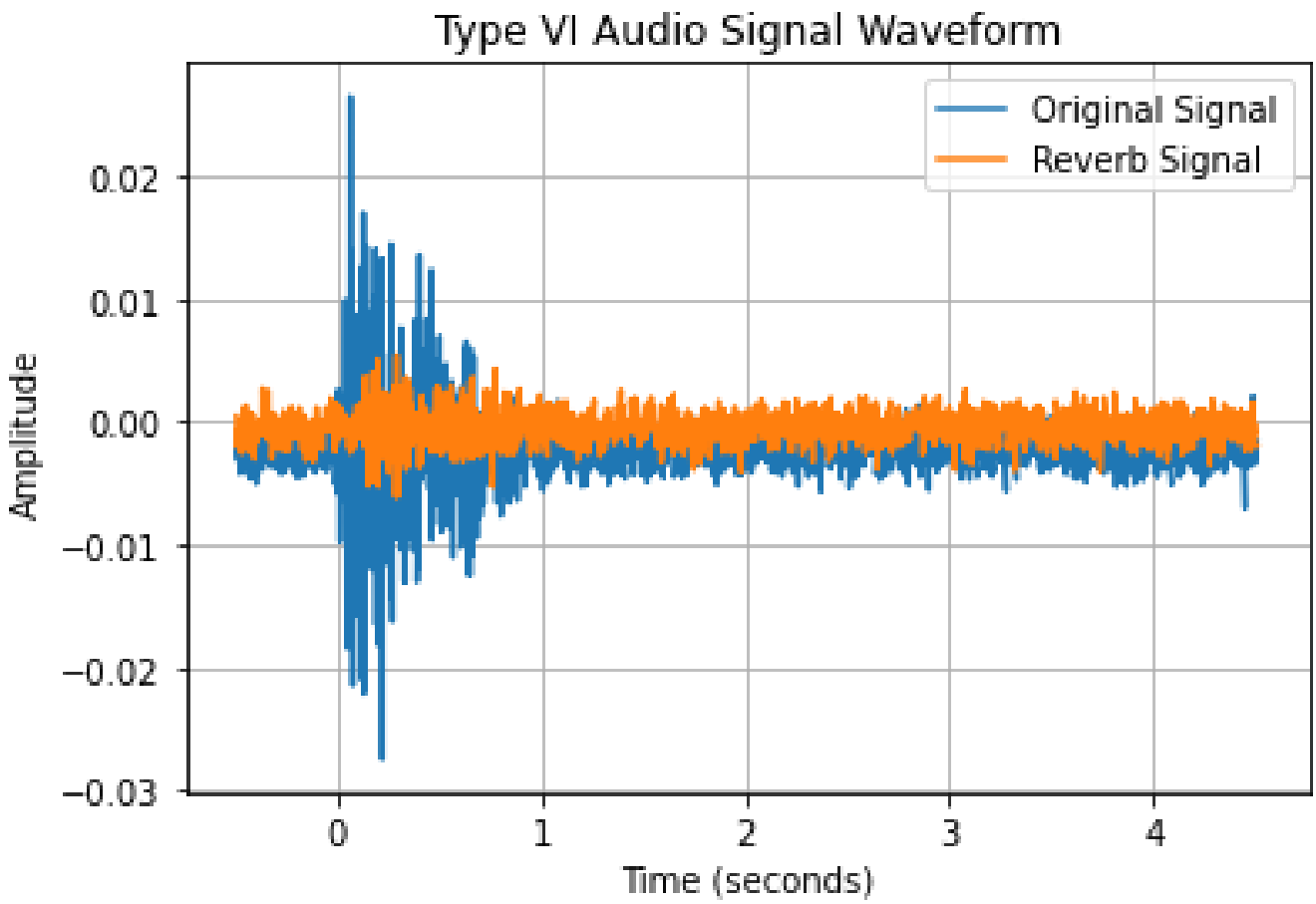


Figure 7: Type II OutC reverb vs. original Drum Darkness Sound effect

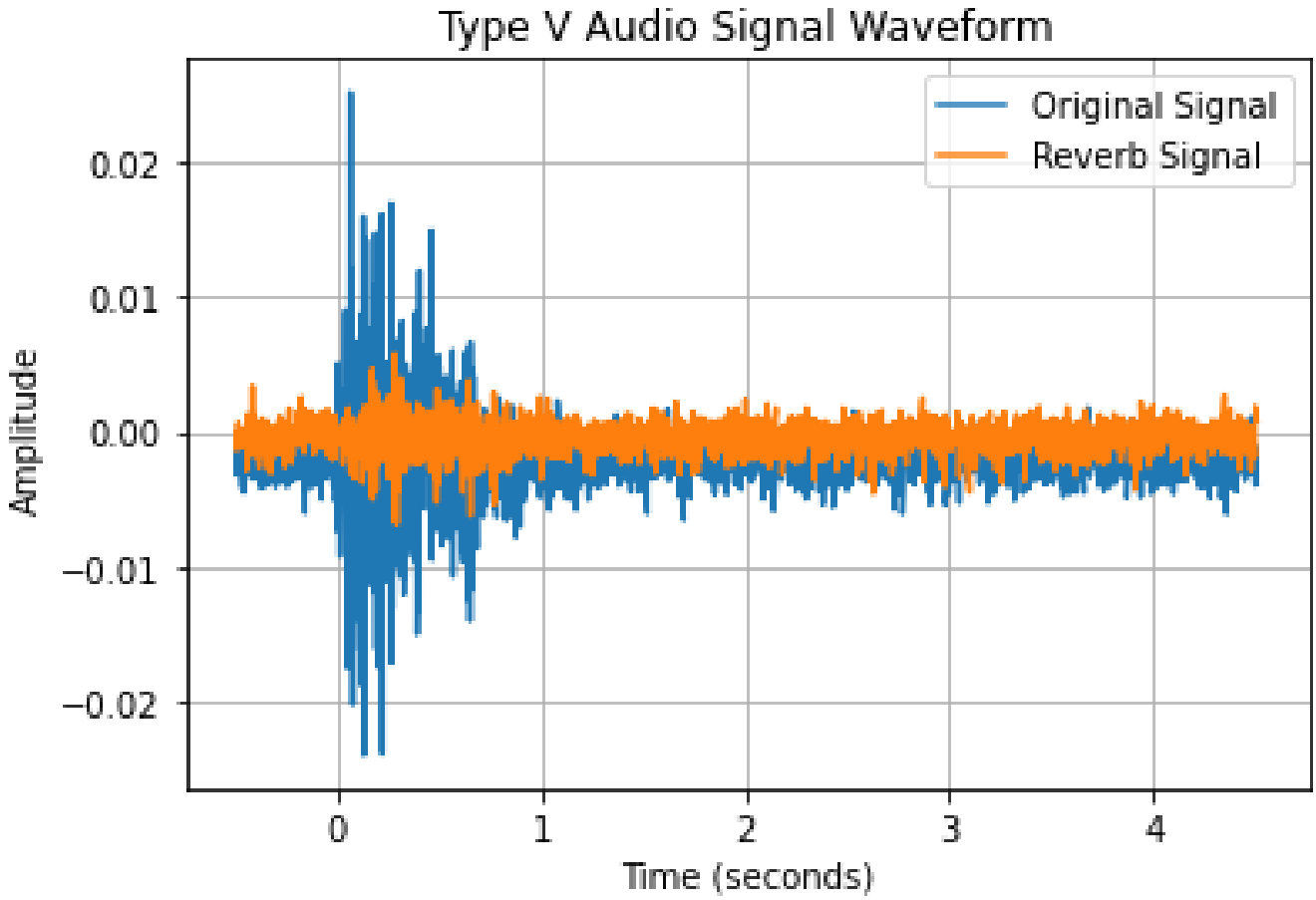


Figure 8: Type II OutD reverb vs. original Drum Darkness Sound effect

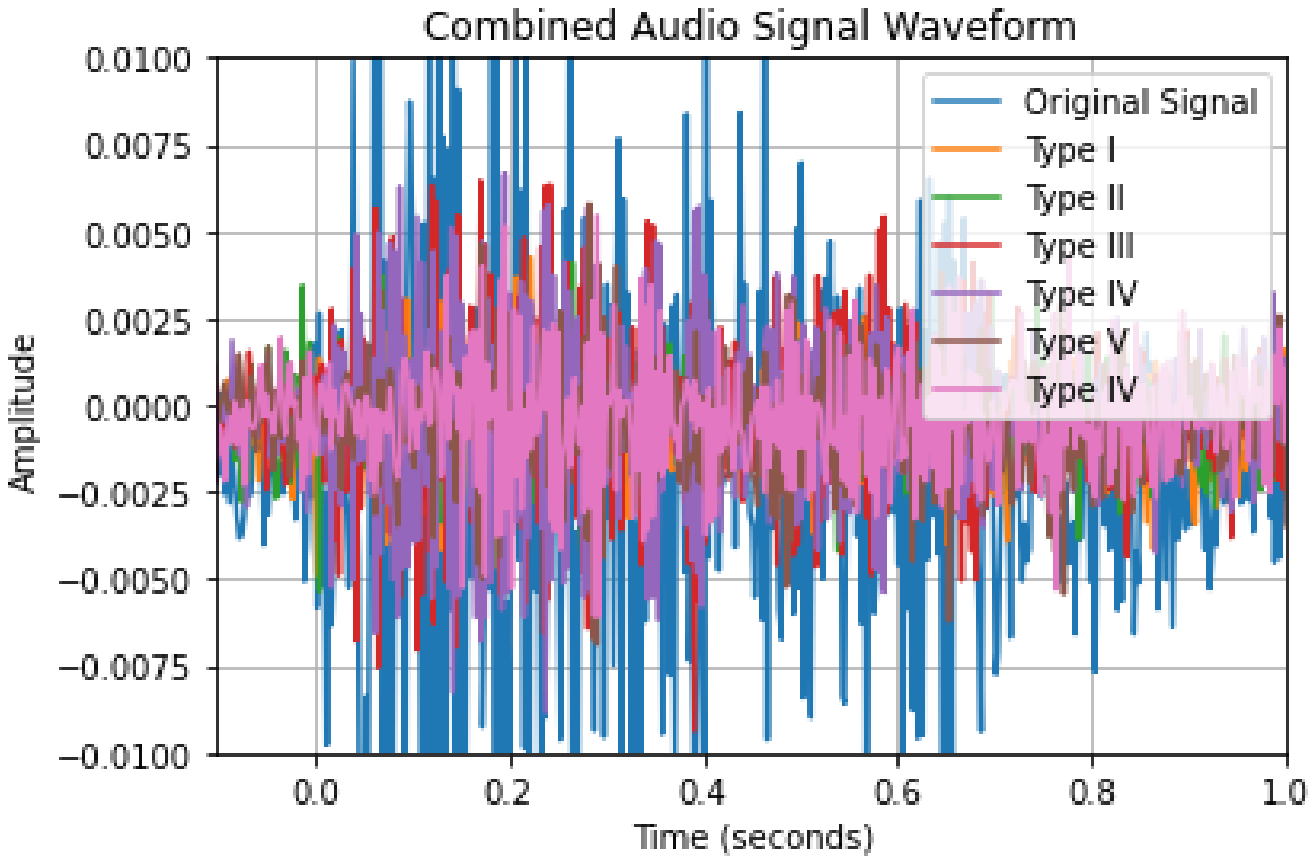


Figure 9: All reverb types vs. original Drum Darkness Sound effect